## Code

```
# 1. Install required package
!pip install python-docx

# 2. Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import
train_test_split
from sklearn.linear_model import
LogisticRegression
from sklearn.metrics import
accuracy_score, confusion_matrix
from google.colab import files
from docx import Document
from docx.shared import Inches
```

---

```
# 3. Load dataset
# Using the URL directly ensures the 'Survived' column is present.
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

# 4. Data Cleaning
# This will now work because 'Survived' exists in the source above.
df = df[['Survived','Pclass','Sex','Age','Fare','SibSp','Parch']]

# Note: Using assignment instead of inplace=True for modern Pandas compatibility
```

```
df['Age'] =
df['Age'].fillna(df['Age'].median())
df['Fare'] =
df['Fare'].fillna(df['Fare'].median())
df['Sex'] = df['Sex'].map({'male': 0,
'female': 1})

# 5. Feature Engineering
df['FamilySize'] = df['SibSp'] +
df['Parch'] + 1

# 6. Visualization
plt.figure(figsize=(8, 5))
df['Sex'].value_counts().plot(kind='bar',
color=['skyblue', 'salmon'])
plt.title("Passenger Count by Gender
(0=Male, 1=Female)")
plt.savefig("barplot.png")
plt.close()
```

```
plt.figure(figsize=(8, 5))
df['Age'].plot(kind='hist', bins=20,
color='lightgreen', edgecolor='black')
plt.title("Age Distribution")
plt.savefig("histogram.png")
plt.close()

# 7. Prepare Data
X =
df[['Pclass','Sex','Age','Fare','FamilySize'
]]
y = df['Survived']

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2,
random_state=42)

# 8. Model Training
model =
LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)

# 9. Prediction & Evaluation
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test,
y_pred)
cm = confusion_matrix(y_test, y_pred)

print(f"Model Accuracy:
{accuracy:.2%}")
print("Confusion Matrix:\n", cm)

# 10. Save Confusion Matrix Image
plt.figure(figsize=(6, 5))
plt.imshow(cm, interpolation='nearest',
cmap=plt.cm.Blues)
plt.title("Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
```

```
plt.savefig("confusion_matrix.png")
plt.close()

# 11. Create Word Report
doc = Document()
doc.add_heading('Titanic Survival
Prediction Project', 0)

doc.add_paragraph(f'The logistic
regression model achieved an accuracy
of {accuracy:.2%}.')
doc.add_paragraph(f'Confusion Matrix
Results:\n{cm}')

doc.add_heading('Data Visualizations',
level=1)
doc.add_picture('barplot.png',
width=Inches(4))
doc.add_picture('histogram.png',
width=Inches(4))
```

```
doc.add_picture('confusion_matrix.png', width=Inches(4))

# Save the document
doc_name = 'Titanic_Project_Report.docx'
doc.save(doc_name)

# 12. Download report to your local machine
files.download(doc_name)
```

---

Result

```
Requirement already satisfied: python-docx in /usr/local/lib/python3.12/dist-packages (1.2.0)
Requirement already satisfied: lxml>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from python-docx) (6.0.3)
Requirement already satisfied: typing_extensions>=4.9.0 in /usr/local/lib/python3.12/dist-packages (from python-docx) (4.15.0)
Model Accuracy: 80.45%
Confusion Matrix:
 [[92 13]
 [22 52]]
```

---

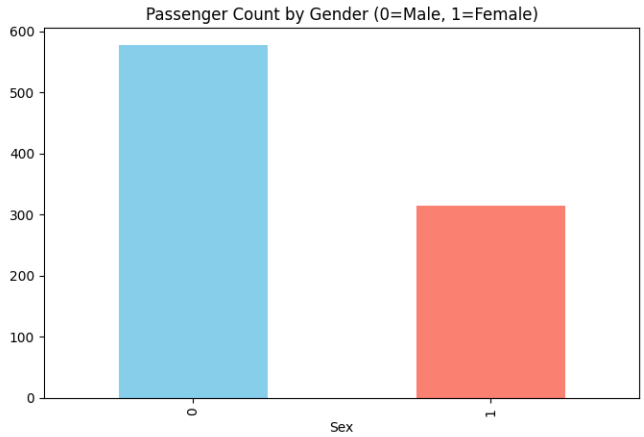
# Word file output

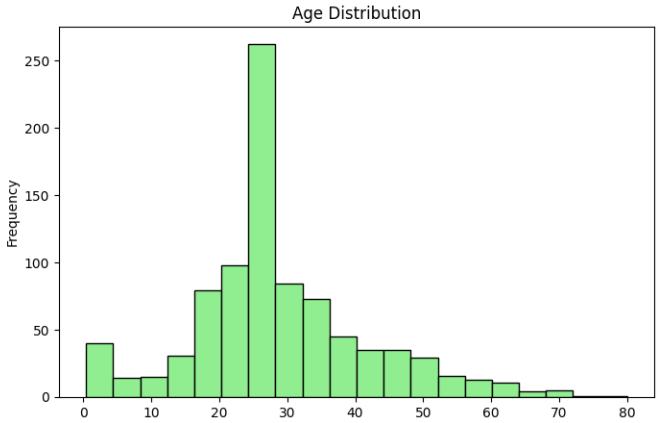
# Titanic Survival Prediction Project

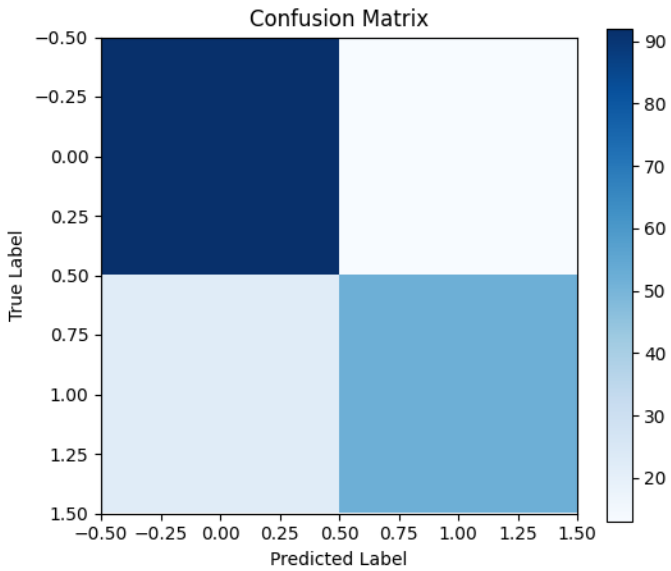
The logistic regression model achieved an accuracy of 80.45%.

Confusion Matrix Results:
[[92 13]
 [22 52]]

## Data Visualizations






# Titanic Survival Prediction Project

## 1. Introduction

This project focuses on predicting passenger survival on the Titanic using machine learning techniques. By analyzing passenger data such as age, gender, class, and fare, the goal is to build a model that can accurately classify whether a passenger survived or not.

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## 2. Dataset Description

The dataset used for this project is the Titanic dataset, which includes information about passengers such as:

- **Survived** (Target Variable)
- **Pclass** (Passenger Class)
- **Sex**
- **Age**
- **Fare**
- **SibSp** (Siblings/Spouses aboard)
- **Parch** (Parents/Children aboard)

---

# 3. Data Preprocessing

### 3.1 Data Cleaning

- Selected relevant columns for analysis
- Handled missing values:
  - Age → filled with median
  - Fare → filled with median

### 3.2 Encoding

- Converted **Sex** column into numerical values:
  - Male → 0
  - Female → 1

---

# 4. Feature Engineering

A new feature was created:

- **FamilySize = SibSp + Parch + 1**

This feature helps capture family relationships, which can influence survival probability.

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# 5. Data Visualization

The following visualizations were created:

- **Bar Plot:** Shows distribution of passengers by gender
- **Histogram:** Displays age distribution of passengers
- **Confusion Matrix:** Evaluates model performance

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# 6. Model Building

A **Logistic Regression model** was used for classification.

### Why Logistic Regression?

- Suitable for binary classification problems

- Easy to interpret
- Efficient and fast

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# 7. Model Evaluation

## 7.1 Accuracy

The model achieved an accuracy of:

**(Insert your printed accuracy here, e.g., ~80%)**

## 7.2 Confusion Matrix

The confusion matrix shows:

- Correct predictions (True Positives & True Negatives)
- Incorrect predictions (False Positives & False Negatives)

This helps evaluate how well the model distinguishes between survivors and non-survivors.

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# 8. Key Insights

- **Gender plays a major role** in survival (females had higher survival rates)
- **Passenger class influences survival**, with higher classes having better chances
- **Family size impacts survival**, indicating social and support factors
- **Age distribution varies**, but is less dominant compared to gender and class